

Brandon Arai

[Available upon request] | [Available upon request] | North Vancouver, BC | Personal Website | Github

Work Availability - Available for 12 month co-op term

Technical Skills

- Languages: Python, C, C#
- Frameworks / Libraries: Pandas, NumPy, Matplotlib, OpenCV
- Tools & Platforms: Git, GitHub, PyCharm, Visual Studio, Unity

Technical Project

Cyber Barrage - GitHub - Itch.io Feb - Jun 2025

- Developed a Unity C# gameplay system using object-oriented programming to separate player control, weapon logic, enemy behavior, and game-state handling into organized scripts.
- Implemented and debugged real-time system mechanics such as shooting, collision detection, health tracking, enemy interactions, and scene progression to improve system reliability.
- Used Unity Editor and Visual Studio debugging tools to identify runtime errors, inspect component references, and resolve logic issues across multiple connected object-oriented scripts.

Chaport Customer Service Chatbot - GitHub May - Jun 2025

- Developed a Python Chatbot API, internal software that requests and replies to local business' embedded Chaport web chat widget for fast and accurate responses.
- Replied by automated posting a clear and informative message through the Hugging Face Mistral AI access token with a series of search functionalities in a local business Excel sheet and website data.
- Built an internal tool to import, extract, and filter out certain keywords and names in a local business' inventory Excel dataset that has over 58,000 rows and 17 columns for the chatbot's info base.
- Maintained a structured Git environment through a path.py and gitignore to avoid pushing API keys, access tokens, and extremely large datasets.

Adaptive Personalized Expression Recognition - Github Feb - Apr 2025

CMPT 419 - Special Topics in Artificial Intelligence, SFU

- Programmed a successful real-time personalized expression recognition system while collaborating alongside a team of 3 people.
- Leveraged Python with PyTorch's Torch and Torchvision packages to train a convolutional neural network on a publicly available FER-2013 dataset with 3,589 examples and our 20 volunteers' unique face features.
- Developed an intuitive PyQt5 and OpenCV user-facing interface for external volunteer users to collect facial signals.
- Accurately displayed the names of users' live facial features from feedback usability surveys, which led to a high project grade.

Champion Data - GitHub Feb - Jun 2024

CMPT 353 - Computational Data Science, SFU

- Implemented a Python-based REST API pipeline for collecting large-scale competitive game statistics for "League of Legends" from Riot Games.
- Automated the filtering and transforming of a 460490-row JSON dataset through Python's Pandas and Numpy to surface actionable performance metrics.
- Supported project GitHub teamwork by coordinating team commits, merges, presenting statistical findings, and clearly communicating debugging decisions across scripts, analysis, and visualization tasks.
- Developed Python data analysis and visualization scripts with Matplotlib plots to identify performance trends, debug statistical patterns, and support data-driven optimization workflows.

Education

BSc Computing Science Sep 2023 - Apr 2027

- Canada

Associate of Science Sep 2021 - Apr 2023

- Canada